



2 Real Analysis 2023

Tutorial 12

A/Prof Elena Berdysheva

Dr Francesco Russo



Uniform continuity

Problem 51 Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$f(x) = \begin{cases} -1, & x \in (-\infty, 0), \\ \frac{1}{2-x}, & x \in [0, 2), \\ 0, & x \in [2, \infty). \end{cases}$$

Prove the following statements. You may prove them by definition or using known statements from the lecture.

- (a) f is not continuous on the interval $[-1, 1]$,
- (b) f is uniformly continuous on the interval $[0, \frac{3}{2}]$,
- (c) f is continuous, but not uniformly continuous on the interval $(0, 2)$.

Solution.

- (a) f is not continuous on the interval $[-1, 1]$, because it is not continuous at $x = 0$. We can show this, for example, as follows. Consider the sequence $x_n = -\frac{1}{n}$, $n \in \mathbb{N}$. Then $\lim_{n \rightarrow \infty} x_n = 0$, but $\lim_{n \rightarrow \infty} f(x_n) = \lim_{n \rightarrow \infty} (-1) = -1 \neq \frac{1}{2} = f(0)$. By Corollary 4.18, f is not continuous at $x = 0$.
- (b) f is continuous on the interval $[0, \frac{3}{2}]$ as a composition of continuous functions. The interval $[0, \frac{3}{2}]$ is compact, since it is closed and bounded. By Theorem 4.30, any function that is continuous on a compact is uniformly continuous there. We conclude that f is uniformly continuous $[0, \frac{3}{2}]$.
- (c) f is continuous on the interval $(0, 2)$ as a composition of continuous functions. To see that f is not uniformly continuous on $(0, 2)$, consider, for example, the sequences $x_n = 2 - \frac{1}{n}$, $y_n = 2 - \frac{1}{n+1}$, $n \in \mathbb{N}$. Then

$$\lim_{n \rightarrow \infty} (x_n - y_n) = \lim_{n \rightarrow \infty} \left(-\frac{1}{n} + \frac{1}{n+1} \right) = 0.$$

On the other hand, $f(x_n) = n$, $f(y_n) = n+1$, and $|f(x_n) - f(y_n)| = 1$ for all $n \in \mathbb{N}$. By Proposition 4.28, f is not uniformly continuous on $(0, 2)$.

Problem 52 (a) Use the Mean Value Theorem to prove that $|\sin x - \sin y| \leq |x - y|$ for any $x, y \in \mathbb{R}$.

(b) Show that the functions $f(x) = x$ and $g(x) = \sin x$ are both uniformly continuous on $\mathbb{R}$.

(c) Show that the function $h(x) = x \sin x$ is not uniformly continuous on $\mathbb{R}$.

Solution.

(a) Notice that $(\sin x)' = \cos x$. By the Mean Value Theorem, for any $x, y \in \mathbb{R}$ there exists θ between x and y such that $\sin x - \sin y = \cos \theta(x - y)$. We obtain that

$$|\sin x - \sin y| \leq |\cos \theta| |x - y| \leq |x - y|.$$

(b) Given $\varepsilon > 0$, take $\delta = \varepsilon > 0$. Then for all $x, y \in \mathbb{R}$

$$\begin{aligned} |x - y| < \delta &\implies |f(x) - f(y)| = |x - y| < \varepsilon, \\ |g(x) - g(y)| &= |\sin x - \sin y| \leq |x - y| < \varepsilon, \end{aligned}$$

as desired.

(c) Consider the sequences $x_n = 2\pi n + \frac{1}{n}$, $y_n = 2\pi n$, $n \in \mathbb{N}$. Then

$$x_n - y_n = \frac{1}{n} \rightarrow 0, \quad n \rightarrow \infty.$$

We have

$$\begin{aligned} h(x_n) &= \left(2\pi n + \frac{1}{n}\right) \sin\left(2\pi n + \frac{1}{n}\right) = \left(2\pi n + \frac{1}{n}\right) \sin \frac{1}{n}, \\ h(y_n) &= 2\pi n \sin(2\pi n) = 0, \end{aligned}$$

$$h(x_n) - h(y_n) = \left(2\pi n + \frac{1}{n}\right) \sin \frac{1}{n} = 2\pi n \sin \frac{1}{n} + \frac{1}{n} \sin \frac{1}{n} = 2\pi \frac{\sin 1/n}{1/n} + \frac{1}{n} \sin \frac{1}{n}.$$

Notice that $\left|\frac{1}{n} \sin \frac{1}{n}\right| \leq \frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$, and consequently $\lim_{n \rightarrow \infty} \frac{1}{n} \sin \frac{1}{n} = 0$.

We know that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$. It follows that $\lim_{n \rightarrow \infty} \frac{\sin 1/n}{1/n} = 1$. Thus,

$$\lim_{n \rightarrow \infty} (h(x_n) - h(y_n)) = 2\pi \neq 0.$$

In particular, we have $|h(x_n) - h(y_n)| \geq \pi$ for n large enough. By Proposition 4.28, h is not uniformly continuous on $\mathbb{R}$.

Uniform convergence

Problem 53 Consider the sequence of functions $f_n : \mathbb{R} \rightarrow \mathbb{R}$, $f_n(x) = x^n(1 - x)$, $n \in \mathbb{N}$.

- (a) Determine the pointwise limit f of the sequence $(f_n)_{n=1}^\infty$ on the interval $[0, 1]$.
- (b) For each $n \in \mathbb{N}$, find the maximum value of f_n on $[0, 1]$.
- (c) Does $(f_n)_{n=1}^\infty$ converge to f uniformly on $[0, 1]$?

Solution.

- (a) We claim that $f(x) = \lim_{n \rightarrow \infty} f_n(x) = 0$ for all $x \in [0, 1]$.

Indeed, if $x = 0$, then $f_n(0) = 0$ for all $n \in \mathbb{N}$, and $f(0) = \lim_{n \rightarrow \infty} f_n(0) = 0$.

Similarly, if $x = 1$, then $f_n(1) = 0$ for all $n \in \mathbb{N}$, and $f(1) = \lim_{n \rightarrow \infty} f_n(1) = 0$.

For a fixed $x \in (0, 1)$, we have that $\lim_{n \rightarrow \infty} x^n = 0$, and therefore $f(x) = \lim_{n \rightarrow \infty} x^n(1 - x) = 0$.

- (b) To determine local extrema of f_n in the interval $(0, 1)$, we consider

$$f'_n(x) = nx^{n-1}(1 - x) - x^n = x^{n-1}(n(1 - x) - x) = x^{n-1}(n - (n + 1)x).$$

We see that $f'_n(x) = 0$ for $x \in (0, 1)$ if and only if $x = \frac{n}{n+1}$. Moreover, f'_n changes its sign from $+$ to $-$ at $x = \frac{n}{n+1}$. It follows that $x = \frac{n}{n+1}$ is a local maximum. Taking into account the values $f_n(0) = f_n(1) = 0$, we conclude that

$$\max_{x \in [0, 1]} f_n(x) = f_n\left(\frac{n}{n+1}\right) = \left(\frac{n}{n+1}\right)^n \frac{1}{n+1}.$$

- (c) Notice that

$$\lim_{n \rightarrow \infty} \left(\frac{n}{n+1}\right)^n \frac{1}{n+1} = \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n}\right)^n} \lim_{n \rightarrow \infty} \frac{1}{n+1} = \frac{1}{e} \cdot 0 = 0.$$

For all $x \in [0, 1]$ we have

$$|f_n(x) - f(x)| = f_n(x) \leq \max_{x \in [0, 1]} f_n(x) = \left(\frac{n}{n+1}\right)^n \frac{1}{n+1} \rightarrow 0, \quad n \rightarrow \infty.$$

The estimate on the right-hand side does not depend on x . It follows that the convergence is uniform on $[0, 1]$.

Problem 54 Consider the sequence of functions $g_n : [0, \infty) \rightarrow \mathbb{R}$, $g_n(x) = \frac{nx}{1+nx}$, $n \in \mathbb{N}$.

- (a) Determine the pointwise limit g of the sequence $(g_n)_{n=1}^\infty$ on $[0, \infty)$.
- (b) Show that $(g_n)_{n=1}^\infty$ converges to g uniformly on $[1, \infty)$.

- (c) Show that $(g_n)_{n=1}^\infty$ does not converge to g uniformly on $[0, \infty)$.

Solution.

- (a) If $x = 0$, then $g_n(0) = 0$ for all $n \in \mathbb{N}$, and $g(0) = \lim_{n \rightarrow \infty} g_n(0) = 0$. If $x > 0$, then

$$g(x) = \lim_{n \rightarrow \infty} \frac{nx}{1 + nx} = \lim_{n \rightarrow \infty} \frac{x}{\frac{1}{n} + x} = 1.$$

The pointwise limit is

$$g(x) = \begin{cases} 0, & x = 0, \\ 1, & x \in (0, \infty). \end{cases}$$

- (b) Let $x \in [1, \infty)$, then

$$|g_n(x) - g(x)| = \left| \frac{nx}{1 + nx} - 1 \right| = \frac{1}{1 + nx} \leq \frac{1}{1 + n} \rightarrow 0, \quad n \rightarrow \infty.$$

The right-hand side does not depend on x . It follows that the convergence is uniform on $[1, \infty)$

- (c) The convergence is not uniform on $[0, \infty)$. Indeed, all functions g_n are continuous on $[0, \infty)$. By Theorem 6.7, a uniform limit of a sequence of continuous functions is continuous. But in our case the limit function g is discontinuous on $[0, \infty)$. It follows that the convergence is not uniform.

Problem 55 Consider the sequence of functions $h_n : \mathbb{R} \rightarrow \mathbb{R}$, $h_n(x) = e^{-nx}$, $n \in \mathbb{N}$.

- (a) Determine the pointwise limit h of the sequence $(h_n)_{n=1}^\infty$ on $[0, \infty)$.
(b) Does $(h_n)_{n=1}^\infty$ converge to h uniformly on $[0, 1]$?

Solution.

- (a) If $x = 0$, then $\lim_{n \rightarrow \infty} h_n(x) = \lim_{n \rightarrow \infty} e^0 = 1$. If $x \in (0, \infty)$, then $e^x \in (0, 1)$, and $\lim_{n \rightarrow \infty} h_n(x) = \lim_{n \rightarrow \infty} (e^{-x})^n = 0$. The pointwise limit is

$$h(x) = \begin{cases} 1, & x = 0, \\ 0, & x \in (0, \infty). \end{cases}$$

- (b) The convergence is not uniform on $[0, 1]$. Indeed, all functions h_n are continuous on $[0, 1]$. By Theorem 6.7, a uniform limit of a sequence of continuous functions is continuous. But in our case the limit function h is discontinuous on $[0, 1]$. It follows that the convergence is not uniform.

Series of functions

Problem 56 Show that the series $\sum_{n=1}^{\infty} \frac{1}{x^2+n^2}$ converges absolutely and uniformly on $\mathbb{R}$.

Solution.

We apply the Weierstrass M-Test. We each $n \in \mathbb{N}$ we have

$$\frac{1}{x^2 + n^2} \leq \frac{1}{n^2} = M_n.$$

The series $\sum_{n=1}^{\infty} M_n = \sum_{n=1}^{\infty} \frac{1}{n^2}$ converges. By the Weierstrass M-Test, the series $\sum_{n=1}^{\infty} \frac{1}{x^2+n^2}$ converges absolutely and uniformly on $\mathbb{R}$.

Problem 57 Show that for any $r > 0$ the series $\sum_{n=1}^{\infty} \frac{1}{x^2 n^2}$ converges absolutely and uniformly on the set $\{x \in \mathbb{R} : |x| \geq r\}$.

Solution.

We apply the Weierstrass M-Test. If $|x| \geq r$, then $\frac{1}{x^2} \leq \frac{1}{r^2}$. We each $n \in \mathbb{N}$ we have

$$\frac{1}{x^2 n^2} \leq \frac{1}{r^2 n^2} = M_n.$$

The series $\sum_{n=1}^{\infty} M_n = \sum_{n=1}^{\infty} \frac{1}{r^2 n^2}$ converges. By the Weierstrass M-Test, the series $\sum_{n=1}^{\infty} \frac{1}{x^2 n^2}$ converges absolutely and uniformly on the set $\{x \in \mathbb{R} : |x| \geq r\}$.

Problem 58 Show that for any $r > 0$ the series $\sum_{n=1}^{\infty} \sin\left(\frac{x}{n^2}\right)$ converges absolutely and uniformly on the interval $[-r, r]$.

Solution.

We apply the Weierstrass M-Test. If $|x| \leq r$ and $n \in \mathbb{N}$, then

$$\left| \sin\left(\frac{x}{n^2}\right) \right| \leq \left| \frac{x}{n^2} \right| \leq \frac{r}{n^2} = M_n.$$

The series $\sum_{n=1}^{\infty} M_n = \sum_{n=1}^{\infty} \frac{r}{n^2}$ converges. By the Weierstrass M-Test, the series $\sum_{n=1}^{\infty} \sin\left(\frac{x}{n^2}\right)$ converges absolutely and uniformly on $[-r, r]$.

Problem 59 Show that the series $\sum_{n=0}^{\infty} \frac{1}{1+x^n}$ does not converge for any $x \in (-1, 1]$, but converges on $(1, \infty)$.

Solution.

If $x \in (-1, 1)$ then $x^n \rightarrow 0$ as $n \rightarrow \infty$, and consequently $\frac{1}{1+x^n} \rightarrow 1$ as $n \rightarrow \infty$. If $x = 1$, then $x^n = 1$ for all $n \in \mathbb{N}$ and $\frac{1}{1+x^n} \rightarrow \frac{1}{2}$ as $n \rightarrow \infty$. We see that the term $\frac{1}{1+x^n}$ does not converge to zero for $x \in (-1, 1]$. It follows that the series diverges.

Notice that the terms $\frac{1}{1+x^n}$ are not defined if $x = -1$ and n is odd.

Now assume that $x \in (1, \infty)$. Then $\frac{1}{1+x^n} \leq \frac{1}{x^n} = \left(\frac{1}{x}\right)^n$, and the series $\sum_{n=0}^{\infty} \left(\frac{1}{x}\right)^n$ converges, since $\left|\frac{1}{x}\right| < 1$. By the Comparison Test, the series $\sum_{n=0}^{\infty} \frac{1}{1+x^n}$ converges.

Problem 60 Consider the series $\sum_{n=0}^{\infty} \frac{x^{2n}}{1+x^{2n}}$. Prove or disprove:

- (a) The series converges on $[-1, 1]$.
- (b) The series converges on $(-1, 1)$.
- (c) The series converges on $[2, 3]$.
- (d) The set of points where the series converges is $(-1, 1)$.
- (e) For all values x for which the series converges we have $\sum_{n=0}^{\infty} \frac{x^{2n}}{1+x^{2n}} \geq \frac{1}{2}$.

Solution.

- (a) This is false because the series does not converge for $x = 1$. If $x = 1$, then all its terms are $\frac{1}{2}$ and, in particular, do not tend to zero. Thus, the series diverges when $x = 1$.
- (b) True. If $x \in (-1, 1)$, then

$$0 \leq \frac{x^{2n}}{1+x^{2n}} \leq x^{2n} = (x^2)^n, \quad n \in \mathbb{N},$$

and the series $\sum_{n=0}^{\infty} (x^2)^n$ converges, because $|x^2| < 1$. By the Comparison Test, the series $\sum_{n=0}^{\infty} \frac{x^{2n}}{1+x^{2n}}$ converges.

- (c) False. If $|x| > 1$, then

$$\lim_{n \rightarrow \infty} \frac{x^{2n}}{1+x^{2n}} = \lim_{n \rightarrow \infty} \frac{1}{\frac{1}{x^{2n}} + 1} = 1.$$

The terms do not tend to zero and therefore the series diverges.

- (d) True. We already know by (b) that the series converges for any $x \in (-1, 1)$. In the proof of (c) we have shown that the series diverges for any x with $|x| > 1$, and in (a) we have shown that the series diverges for $x = 1$. Notice that exactly the same proof as in (a) can be used to show divergence at $x = -1$. Hence, the series converges if and only if $x \in (-1, 1)$.
- (e) True. Notice that for $n = 0$ we always have $\frac{x^0}{1+x^0} = \frac{1}{2}$, and for all $n \in \mathbb{N}$ we have $\frac{x^{2n}}{1+x^{2n}} \geq 0$. Thus,

$$\sum_{n=0}^{\infty} \frac{x^{2n}}{1+x^{2n}} \geq \frac{x^0}{1+x^0} = \frac{1}{2}.$$

Power series

Problem 61 For each of the following series, determine its radius of convergence.

$$(a) \sum_{n=0}^{\infty} \frac{x^n}{n^n}, \quad (b) \sum_{n=0}^{\infty} \frac{n^n x^n}{n!}, \quad (c) \sum_{n=0}^{\infty} \frac{(n!)^2 x^n}{(2n)!}.$$

Solution.

(a) We have

$$\frac{1}{R} = \lim_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left(\frac{1}{n^n} \right)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

It follows that $R = \infty$.

(b) We have

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{(n+1)^{n+1}}{(n+1)!} \frac{n!}{n^n} = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e.$$

It follows that $R = \frac{1}{e}$.

(c) We have

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{((n+1)!)^2 (2n)!}{(2n+2)! (n!)^2} = \lim_{n \rightarrow \infty} \frac{(n+1)^2}{(2n+2)(2n+1)} = \frac{1}{4}.$$

It follows that $R = 4$.